

2020 HKDSE Physics

Paper 2 — Elective Sections

Editable transcription of all questions and formulae. Only necessary, non-duplicated original diagrams are inserted. Each official answer is followed by a verified concise full-mark solution in English, with editable subscripts and superscripts.

Editable transcription with official answers and verified concise full-mark reasoning in English.

Question 1

Section A: Astronomy and Space Science

Multiple-choice questions

Official answers are placed directly below each question.

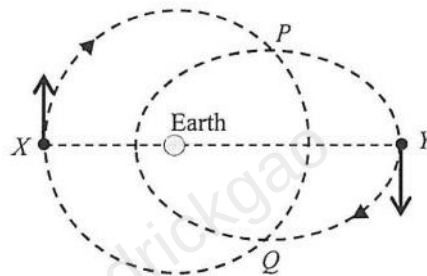
1.1 Which of the following represents the arrangement in descending order of size?

- A. star cluster > galaxy > planetary system
- B. star cluster > planetary system > galaxy
- C. galaxy > planetary system > star cluster
- D. galaxy > star cluster > planetary system

Official answer: D

A galaxy is larger than a star cluster, and a star cluster is larger than a planetary system. Answer: D.

1.2 Two satellites X and Y are revolving in a clockwise direction about the Earth as shown. The diameter of the circular orbit of X equals the length of the major axis of the elliptical orbit of Y. The two orbits intersect at P and Q. At the instant shown, the two satellites and the Earth are on a straight line. Which deductions are correct?



(1) X and Y have the same acceleration when they pass P.

- (2) The speed of X is greater than that of Y at the instant shown.
 (3) The satellites will not meet each other at P or at Q.

- A. (1) and (2) only
 B. (1) and (3) only
 C. (2) and (3) only
 D. (1), (2) and (3)

Official answer: D

The circular orbit and the ellipse have the same semi-major axis, so their periods are equal. At P their distances from the Earth are equal, so their gravitational accelerations are equal. At the instant shown, Y is farther from the Earth than X and therefore has the lower speed. Their equal periods and shown phase relation also prevent a meeting at P or Q. Thus all three statements are correct. Answer: D.

1.3 The mass of Mars is 0.107 times that of the Earth. The radius of Mars is 0.532 times that of the Earth. What is the escape velocity of Mars in terms of the escape velocity of the Earth v_E ?

- A. $0.201v_E$
 B. $0.378v_E$
 C. $0.449v_E$
 D. $0.615v_E$

Official answer: C

Escape speed is proportional to $\sqrt{(M/R)}$. Therefore $v_N/v_E = \sqrt{(0.107/0.532)} = 0.449$. Answer: C.

1.4 The luminosity of star P is double that of star Q. The brightness of P is eight times that of Q. What can be deduced about the respective distances of stars P and Q from the Earth?

- A. P's distance from the Earth is 2 times that of Q.
- B. Q's distance from the Earth is 2 times that of P.**
- C. P's distance from the Earth is 4 times that of Q.
- D. Q's distance from the Earth is 4 times that of P.

Official answer: B

Brightness $b = L/(4\pi d^2)$. With $b_P/b_Q = 8$ and $L_P/L_Q = 2$, $8 = 2(d_Q/d_P)^2$, so $d_Q = 2d_P$. Answer: B.

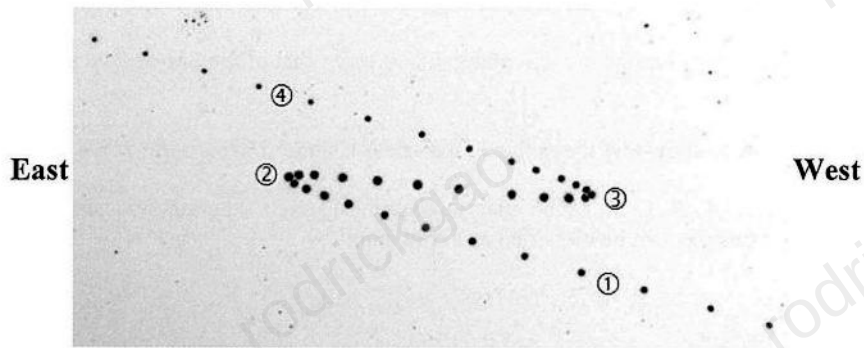
1.5 The edge-on view of a binary system shows that the wavelength of the calcium K spectral line of one of the stars changes by ± 0.3 nm from the same line ($\lambda = 393.4$ nm) obtained in a laboratory. The period of the star is 69 hours. Find the radius of its orbit.

- A. 2.5×10^9 m
- B. 1.5×10^9 m
- C. 9.0×10^9 m**
- D. 5.6×10^{10} m

Official answer: C

The Doppler shift gives $v/c = \Delta\lambda/\lambda = 0.30/393.4$, so $v = 2.29 \times 10^5$ m s⁻¹. With period 69 h, $r = vT/(2\pi) = 9.0 \times 10^9$ m. Answer: C.

1.6 The figure shows the motion of Mars from bottom right to top left against the background night sky over a period of about 7 months. At which point along the path was Mars closest to the Earth?



- A. somewhere between ① and ②
- B. somewhere between ② and ③
- C. somewhere between ③ and ④
- D. either at turning point ② or at turning point ③

Official answer: B

Mars is closest to Earth near opposition. Its apparent motion reverses during the central retrograde segment between the two stationary points marked 2 and 3. Answer: B.

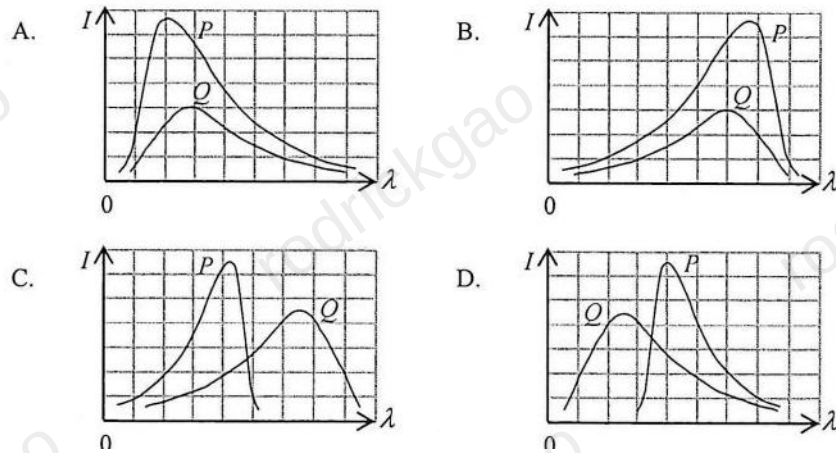
1.7 A star, which is 4.2 light years from the Sun, is observed from the Earth 6 months apart. Estimate the maximum angular difference in the observed positions of this star.

- A. 0.8 arc seconds
- B. 1.3 arc seconds
- C. 1.6 arc seconds
- D. 2.6 arc seconds

Official answer: C

A distance of 4.2 ly is about 1.29 pc, so the annual parallax angle is $p = 1/1.29 = 0.775$ arcsec. Observations six months apart show a displacement $2p \approx 1.55$ arcsec ≈ 1.6 arcsec. Answer: C.

1.8 Star P has a higher surface temperature than star Q. However, star Q has a larger radius. Which graph shows the distribution of spectral intensity I (in W m^{-2} per nm) with wavelength λ (in nm) of the electromagnetic radiation emitted from the surfaces of P and Q? Blackbody radiation is assumed for both stars. The four graph options A–D are reproduced from the original paper below.



Official answer: A

Spectral intensity per unit surface area depends on temperature, not stellar radius. The hotter star P has a shorter peak wavelength and a greater intensity at every wavelength, as in graph A. Answer: A.

Structured question

A certain star X at a distance of about 50 kpc from the Earth exploded a very long time ago and became Supernova 1987A (SN 1987A). The light from this supernova first reached the Earth in 1987.

(a) Estimate how long ago the above explosion took place. Give your answer in years to 3 significant figures. (1 mark)

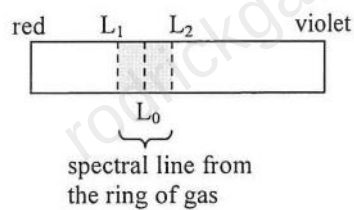
(b) At maximum brightness, SN 1987A has an apparent magnitude of +2.9. Is the absolute magnitude of SN 1987A at maximum brightness smaller than, larger than or equal to +2.9? Explain your answer. (2 marks)

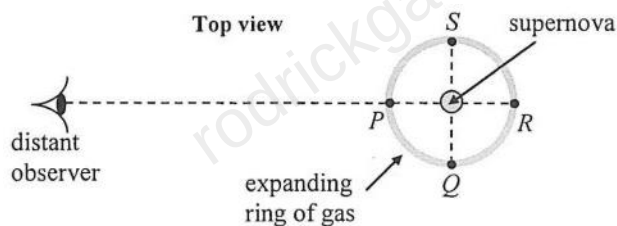
Before the explosion of star X, its luminosity was about 40 000 times that of the Sun and its surface temperature was 3.1 times that of the Sun.

(c) (i) Use Stefan's law to show that the radius of star X is about 20 times the Sun's radius. (2 marks)

(ii) In which region, A, B, C or D, on the Hertzsprung–Russell diagram was star X located? Explain whether or not this star can be classified as a 'red giant'. (2 marks)

(d) A special feature of SN 1987A is that a circular ring of gas surrounds the supernova. The gas was ejected by star X some time before it exploded. As shown in Figure 1.1, each point on this ring is expanding outwards at a constant speed from the supernova. Suppose a distant observer on the plane containing the ring views a certain spectral line from the ring of gas and finds that it covers wavelengths between the limits L_1 and L_2 as shown in Figure 1.1. L_0 is the wavelength of that spectral line when observed in the laboratory. State the respective wavelengths that originate from point Q and point R on the ring. Explain your answer. (3 marks)





Official marking scheme

Solution		Marks	Remarks
1.	(a) Distance 50 kpc = 50000 × 3.26 ly = 163000 ly Thus the explosion of the star took place 163000 years ago. <u>Accepted range: 163000 ~ 164000 years</u> or $50000 \times 3.09 \times 10^{16} / (3 \times 10^8 \times 365 \times 24 \times 3600) = 163305$ or $50000 \times 206265 \times 1.5 \times 10^{11} / (3 \times 10^8 \times 365 \times 24 \times 3600) = 163516$	1A	Note: 2020 – 1987 = 33 years can be ignored.
		1	
	(b) SN 1987A at 50 kpc, being much farther away than 10 pc, would be much brighter (than that corresponding to +2.9) if it were placed at 10 pc . <u>NO MARK awarded without 'placed at 10 pc'</u> Hence (the numerical value of) its absolute magnitude is much smaller than +2.9 / the apparent magnitude.	1A	[Absolute magnitude is the apparent brightness of the object at 10 pc.]
		1A	
		2	
	(c) (i) Take L_S , R_S and T_S to be the luminosity, radius and surface temperature of the Sun respectively, whereas let L_X , R_X and T_X be those of X respectively. By Stefan's law, $L_S = \sigma(4\pi R_S^2)T_S^4$ and $L_X = \sigma(4\pi R_X^2)T_X^4$ Therefore, $\frac{L_X}{L_S} = \left[\frac{R_X}{R_S}\right]^2 \left[\frac{T_X}{T_S}\right]^4$ $40000 = \left[\frac{R_X}{R_S}\right]^2 [3.1]^4$ $\frac{R_X}{R_S} = 20.81165 \approx 20.8$	1M	Note: red giants are in region B, star X is in fact a blue supergiant.
		1M	
		2	
	(ii) Region A. Not a 'red giant' as the temperature of star X is (much) higher than that of the Sun. / Red giants are in region B	1A	
		1A	
		2	
	(d) Q : L_0 <u>or</u> R : L_1 According to Doppler's effect, <u>gas at R receding</u> from the observer gives rise to red shift (vice versa for P), while there is <u>no velocity component</u> for gas at Q (and S) towards / away from the observer, no Doppler / blue / red shift.	1A	
		1A	
		1A	
		3	

(a) 50 kpc = 50 000 pc $\approx 1.63 \times 10^5$ ly, so the explosion occurred about 1.63×10^5 years ago.

(b) If the supernova were only 10 pc away, it would appear far brighter; its absolute magnitude would be numerically much smaller than +2.9.

(c)(i) From $L/L_S = (R_X/R_S)^2(T_X/T_S)^4$, $4.0 \times 10^4 = (R_X/R_S)^2(3.1)^4$, giving $R_X/R_S = 20.8$.

(ii) X lies in region A. It is not a red giant because its surface temperature is much higher; it is a blue supergiant.

(d) Q has no line-of-sight velocity, so it produces L_0 . R is receding and produces the red-shifted line L_1 .

Question 2

Section B: Atomic World

Multiple-choice questions

2.1 Which of the following statements about Rutherford's scattering experiment is/are correct?

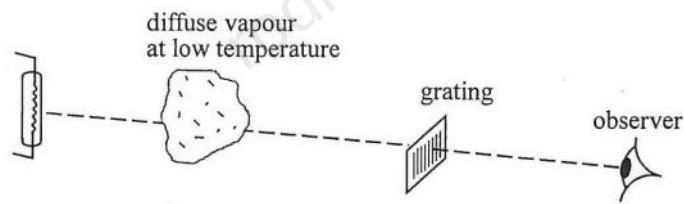
- (1) Thin gold foil was used so that each α particle would most likely be scattered by one gold nucleus only.
 - (2) If β particles were directed towards the gold foil instead of α particles, they could be deflected by both the gold nuclei and the electrons in the foil.
 - (3) Gold foil was used because gold contains free electrons.
- A. (1) only
 - B. (3) only
 - C. (1) and (2) only
 - D. (2) and (3) only

Official answer: C

A thin foil minimizes multiple scattering. β -particles can be deflected by both repulsion from foil electrons and attraction to nuclei. Gold was chosen mainly because it is highly malleable, not because of free electrons. Thus

(1) and (2) are correct. Answer: C.

2.2 In the experiment shown, dark lines are observed on a colour background. Which explanation is correct?



- A. No photons corresponding to the dark lines were emitted from the source.
- B. No photons corresponding to the dark lines were emitted from the vapour.
- C. Photons corresponding to the dark lines from the source were absorbed by the vapour.
- D. Photons corresponding to the dark lines from the source were scattered back to the source by the vapour.

Official answer: C

A hot dense source gives a continuous spectrum. Passing it through cooler vapour removes photons whose energies match atomic transitions, producing dark absorption lines. Answer: C.

2.3 The figure shows the emission lines of the hydrogen spectrum associated with electron transitions from excited energy levels to the ground state ($n = 1$). It is known that the energy level of the ground state is $E_1 = -13.6 \text{ eV}$. Which statement is correct?

$$E_1 = -13.6 \text{ eV.}$$



- A. Spectral line X has the highest frequency.
- B. The shortest wavelength of the lines in this spectral series is about 90 nm.
- C. These spectral lines are in the infra-red region.
- D. Some of these lines also appear in other spectral series of the hydrogen spectrum.

Official answer: B

The Lyman series ends at $n = 1$. Its shortest wavelength is the series limit, for which $E = 13.6 \text{ eV}$, giving $\lambda = hc/E \approx 91.2 \text{ nm}$. Answer: B.

2.4 Which statements about the Bohr atom model are correct?

- (1) It can explain why α particles can be rebounded by a thin gold foil.
- (2) It can give the atomic spectra of a singly ionized helium atom (He^+).
- (3) A postulate of this model is that the angular momentum of the electron in a hydrogen atom is quantized.
- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

Official answer: D

The nuclear model explains large-angle α scattering through a small massive positive nucleus. The Bohr model explains the hydrogen-like spectrum of He^+ and quantizes electron angular momentum. All three statements are correct. Answer: D.

2.5 An atom has three energy levels X, Y and Z. When the atom transits from X to Y, it emits a photon of wavelength λ_1 . The atom transits from Y to Z when it absorbs a photon of wavelength λ_2 , with $\lambda_2 > \lambda_1$. These energy levels arranged in descending order are

- A. Z, X, Y
- B. Z, Y, X
- C. X, Z, Y
- D. X, Y, Z

Official answer: A

$X \rightarrow Y$ emits a photon, so $E_X > E_Y$. $Y \rightarrow Z$ absorbs a shorter-wavelength photon because $\lambda_2 < \lambda_1$, so $E_Z - E_Y > E_X - E_Y$. Hence $E_Z > E_X > E_Y$. Answer: A.

2.6 An electron beam accelerated through a potential difference V is directed to a thin crystal layer. The diffraction pattern obtained is similar to that produced by X-rays of wavelength λ . What potential difference should be used for accelerating the electron beam so as to give a diffraction pattern similar to that produced by X-rays of wavelength 2λ ?

- A. $V/4$
- B. $V/2$
- C. $2V$
- D. $4V$

Official answer: A

For electrons accelerated through V , $\lambda = h/\sqrt{(2meV)}$. Doubling the de Broglie wavelength therefore requires the accelerating voltage to fall to $V/4$. Answer: A.

2.7 Which of the following statements about a scanning tunnelling microscope (STM) is/are correct?

- (1) STM can reveal the internal structure of a specimen as three-dimensional images are produced.
- (2) In STM imaging, the surface of the specimen must be electrically conducting.
- (3) The resolving power of STM is limited by the Rayleigh criterion.

- A. (1) only
- B. (2) only**
- C. (3) only
- D. (1), (2) and (3)

Official answer: B

A scanning tunnelling microscope maps a conducting surface rather than the internal structure of a specimen, and its resolution is not limited by visible-light diffraction. Only statement (2) is correct. Answer: B.

2.8 Which of the following statements about materials in bulk form and in nano size are correct?

- (1) Most materials exhibit different colour in the states mentioned above.
- (2) Most materials in nano size have lower melting point.
- (3) Materials in nano size are usually more efficient in serving as catalysts.

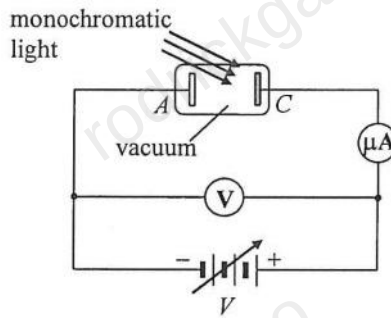
- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)**

Official answer: D

Reducing a material to nanoscale can change its colour, lower its melting point and greatly increase its surface-area-to-volume ratio, improving catalytic activity. All three statements are correct. Answer: D.

Structured question

In order to demonstrate the photoelectric effect, electrodes A and C of the photocell in Figure 2.1 are connected to a potential difference V which can be read from the high-resistance voltmeter. This potential difference can vary from 0 V to 2.5 V. When monochromatic light of wavelength 300 nm is incident on electrode C, the microammeter of negligible internal resistance shows a reading.



- (a) (i) State the part of the electromagnetic spectrum (ultra-violet, blue, green, red or infra-red) to which the incident light belongs. (1 mark)
- (ii) According to wave theory, there should be a 'time delay' for photoelectric emission to occur. However, the experimental result shows that photoelectric emission is immediate. State the implication of such an experimental result. (1 mark)
- (b) The applied potential difference is adjusted until the microammeter reading just falls to zero when $V = 1.7$ V.
- (i) State and explain whether the microammeter reading would change if incident light of the same wavelength but higher intensity is used. (2 marks)
- (ii) Calculate the work function, in eV, of electrode C. (3 marks)
- (c) Now the applied potential difference is adjusted until $V = 0.8$ V and the microammeter registers $0.4 \mu\text{A}$.
- (i) Estimate the number of photoelectrons reaching electrode A in one second. (1 mark)
- (ii) State the maximum kinetic energy, in eV, of the photoelectrons reaching A. Explain why not all photoelectrons reaching A possess this amount of kinetic energy. (2 marks)

Official marking scheme

Q. A (%)	Q. A (%)	I. B (%)	O. D (%)
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Solution	Marks	Remarks
2. (a) (i) ultra-violet (UV)	1A	
	1	
(ii) Light (energy) is (transferred to electrons of the cathode) in packets or (whole number of) quanta (i.e. quantized) called photons.	1A	
	1	
(b) (i) Microammeter reading remains zero, energy E of incident photons remains unchanged, although intensity $\uparrow$ causing more photons incident but no effect on maximum KE of the electrons emitted or on photoemission.	1A	
	1A	
	2	
(ii) Energy of photon = $\frac{hc}{\lambda}$ $= \frac{(6.63 \times 10^{-34})(3 \times 10^8)}{300 \times 10^{-9}}$ $= 6.63 \times 10^{-19} \text{ J}$ $= 4.14375 \text{ (eV)} \approx 4.14 \text{ (eV)}$ Work function $\phi = 4.14 - 1.7$ $= 2.44375 \text{ (eV)} \approx 2.44 \text{ (eV)}$	1M 1A	
	3	
(c) (i) No. of photoelectrons reaching A in 1 s $= \frac{0.4 \times 10^{-6}}{1.6 \times 10^{-19}} = 2.5 \times 10^{12}$	1A	
	1	
(ii) $1.7 - 0.8 = 0.9 \text{ (eV)}$ or $4.14 - 2.4 - 0.8 = 0.94 \text{ (eV)}$. Electrons inside cathode C (not on surface) need an amount of energy more than the work function to escape / be emitted from C.	1A 1A	
	2	

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(a)(i) Ultraviolet radiation is used.

(ii) Electromagnetic energy is transferred in discrete packets, or quanta, called photons.

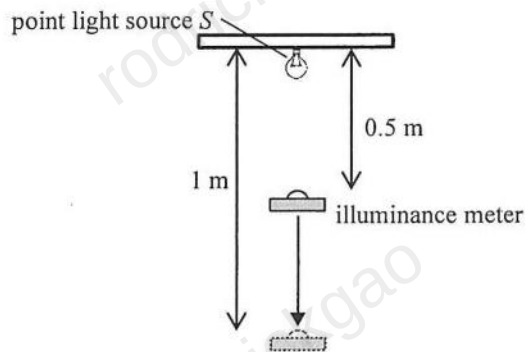
(b)(i) At the stopping potential, the photocurrent remains zero when intensity increases because intensity changes photon number but not photon energy or $K_{\max}$.(ii) The photon energy is $hc/\lambda = 4.14 \text{ eV}$, so the work function is $\phi = 4.14 - 1.70 = 2.44 \text{ eV}$.(c)(i) $N = I/e = (0.40 \times 10^{-6}) / (1.60 \times 10^{-19}) = 2.5 \times 10^{12} \text{ electrons s}^{-1}$.(ii) The maximum kinetic energy on reaching A is $1.70 - 0.80 = 0.90 \text{ eV}$. Electrons starting below the surface lose different amounts of energy while escaping, so they arrive with a range of kinetic energies.

Question 3

Section C: Energy and Use of Energy

Multiple-choice questions

3.1 An illuminance meter is placed 0.5 m directly underneath a point light source S emitting a certain luminous flux as shown. If the luminous flux emitted by S is doubled and the meter is lowered to a position 1 m underneath S, what would be the change in the meter reading?



- A. decreases by 25%
- B. decreases by 50%
- C. remains unchanged
- D. increases by 50%

Official answer: B

Illuminance from a point source is proportional to luminous flux divided by distance squared. Doubling the flux and doubling the distance changes the reading by $2/2^2 = 1/2$, a 50% decrease. Answer: B.

3.2 A solar furnace uses a reflector of area 9 m^2 to collect sunlight so as to heat a piece of iron of mass 2 kg. The efficiency of the furnace is 50%. Estimate the time taken to heat the iron from 30°C to 90°C .

Given: specific heat capacity of iron = $450 \text{ J kg}^{-1} ^\circ\text{C}^{-1}$; solar power per unit area received on the Earth's surface = 1000 W m^{-2} .

- A. 12 s
- B. 24 s
- C. 48 s
- D. 96 s

Official answer: A

The useful input power is $0.5(1000)(9) = 4500 \text{ W}$. The required energy is $mc\Delta T = 2(450)(60) = 54\,000 \text{ J}$, so $t = 54\,000/4500 = 12 \text{ s}$. Answer: A.

3.3 An electric vehicle's battery of capacity 40 kW h is completely discharged initially. It is charged with a terminal voltage of 220 V at an average current of 32 A. Estimate the time required to fully charge this battery. Assume that 20% energy loss occurs during charging.

- A. 4.6 hours
- B. 5.7 hours
- C. 6.8 hours
- D. 7.1 hours

Official answer: D

Storing 40 kW h in the battery at 80% charging efficiency requires $40/0.80 = 50$ kW h of electrical input. The input power is $VI = 220(32) = 7.04$ kW, so $t = 50/7.04 = 7.10$ h. Answer: D.

3.4 What are the reasons for a microwave oven having an 'end-use energy efficiency' of less than 100%?

- (1) Some energy is lost during the transmission of electrical energy from the power plant to the oven.
- (2) The microwave oven cannot convert all the electrical energy to microwave energy.
- (3) Some microwaves are absorbed by the oven's body and do not reach the food.

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

Official answer: C

End-use efficiency compares the electrical energy delivered to the oven with the useful energy transferred to the food, so upstream transmission loss is excluded. Energy lost in the appliance and incomplete useful absorption both reduce the efficiency. Thus (2) and (3) are correct. Answer: C.

3.5 During summertime, heat flows into a house through walls of a certain thickness. The rate of heat flow per unit area of the wall would be reduced if

- (1) the temperature difference between the outer and inner surfaces of the wall is smaller;
- (2) the thickness of the wall is increased;
- (3) material of a larger U-value is used for building the wall.

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

Official answer: A

For a building element, heat-transfer rate per unit area is $U\Delta T$. A smaller temperature difference and a greater insulation thickness reduce the transfer, while a larger U increases it. Thus (1) and (2) are correct. Answer: A.

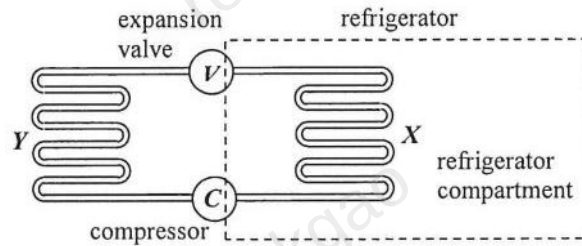
3.6 A room is kept cool by an air-conditioner of cooling capacity P. The temperatures inside and outside the room are 27°C and 31°C respectively. The rates of heat flowing into the room by radiation through windows and by conduction are in the ratio 1: 4. If the cooling capacity is raised to 2P while the temperature outside is still 31°C , estimate the temperature inside the room. Assume that the rate of heat flow into the room by radiation is unchanged.

- A. 21°C
- B. 22°C
- C. 23°C
- D. 25°C

Official answer: B

Initially radiation loss is $P/5$ and conduction loss is $4P/5$ for a 4°C difference. At cooling capacity 2P, radiation is still $P/5$, so conduction must be $9P/5$. Since conduction is proportional to ΔT , the new difference is $(9/4)(4) = 9^{\circ}\text{C}$ and the interior temperature is $31 - 9 = 22^{\circ}\text{C}$. Answer: B.

3.7 A simplified schematic diagram of a refrigerator is shown. What is the direction of flow of the refrigerant through expansion valve V? Which component, X or Y, contains refrigerant at a higher temperature?



- A. direction $X \rightarrow V \rightarrow Y$; higher-temperature component X
- B. direction $X \rightarrow V \rightarrow Y$; higher-temperature component Y
- C. direction $Y \rightarrow V \rightarrow X$; higher-temperature component X
- D. direction $Y \rightarrow V \rightarrow X$; higher-temperature component Y

Official answer: D

The refrigerant flows from the high-pressure condenser side Y through the expansion valve V to the low-pressure evaporator X. The condenser side Y is hotter than X. Answer: D.

3.8 Which of the following is/are renewable energy source(s)?

- (1) wind power
- (2) natural gas
- (3) nuclear power

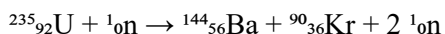
- A. (1) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

Official answer: A

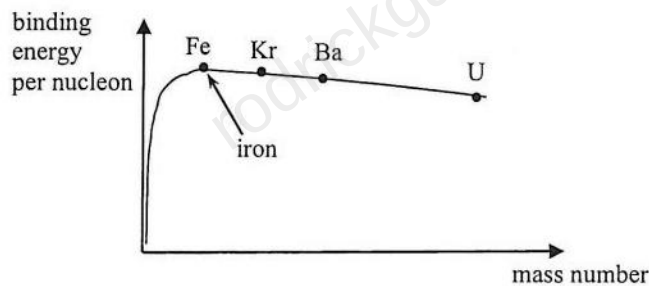
Wind is renewable. Natural gas and nuclear fuel are finite non-renewable energy sources. Answer: A.

Structured question

Pressurized water reactors constitute the majority of the world's nuclear power plants. In the reactor, energy is produced by fission of uranium-235 nuclei ($^{235}_{92}\text{U}$). A typical fission reaction is:



- (a) Referring to the binding energy curve in Figure 3.1, explain why uranium-235 nuclei tend to undergo fission. (2 marks)



(b) The binding energy of a uranium-235 nucleus is 1783 MeV.

(i) What does the above statement mean? (1 mark)

(ii) Find the energy released, in MeV, in the fission of a uranium-235 nucleus.

Given: binding energy per nucleon of $^{144}_{56}\text{Ba}$ nucleus = 8.27 MeV per nucleon; binding energy per nucleon of $^{90}_{36}\text{Kr}$ nucleus = 8.59 MeV per nucleon. (2 marks)

(c) (i) In the reactor of a nuclear power plant, a total energy of 1.30×10^{30} MeV would be released if all uranium-235 nuclei in the fuel rods underwent fission. Given that the mean power output of the power plant is 500 MW and the efficiency in converting nuclear energy to electrical energy is 40%, estimate the time, in years, for which the fuel rods can be used. Take 1 year = 3.15×10^7 s. (2 marks)

(ii) State a reason why the fuel rods are usually replaced before the time estimated in (c)(i) has elapsed. (1 mark)

(d) Explain the role of the following in a fission reactor:

(i) moderator

(ii) control rods (2 marks)

Official marking scheme

3. A (70)	3. B (70)	3. C (70)	3. D (70)
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Solution	Marks	Remarks
3. (a) The fission products / nuclides are with higher <u>binding energy per nucleon</u> than uranium-235. Therefore, energy is released in fission and the resulting nuclides are <u>more stable</u> .	1A 1A 2	1A for comparing the binding energy per nucleon before and after fission of $^{235}_{92}\text{U}$ 1A for more stable after fission
(b) (i) It represents the energy <u>required</u> to separate all nucleons (protons and neutrons) of uranium-235 to <u>infinity</u> / far apart / separate completely.	1A 1	Accept: Energy <u>released</u> by combining protons and neutrons (nucleons) into a single nucleus.
(ii) Binding energy (B.E.) of $^{235}_{92}\text{U}$ nucleus = 1783 MeV B.E. of $^{144}_{56}\text{Ba}$ nucleus = $8.27 \times 144 = 1190.88$ MeV B.E. of $^{90}_{36}\text{Kr}$ nucleus = $8.59 \times 90 = 773.1$ MeV Hence, energy released in fission = $(1190.88 + 773.1) - 1783$ = 180.98 (MeV) ≈ 181 (MeV)	1M 1A 2	1M for calculating the total binding energy of $^{144}_{56}\text{Ba}$ & $^{90}_{36}\text{Kr}$ from the binding energy per nucleon Accept: $180 - 181$ MeV $(2.88 - 2.90) \times 10^{-11}$ J
(c) (i) <u>Total energy released $\times$ efficiency</u> Power output = $\frac{(1.30 \times 10^{30} \times 10^6)(1.6 \times 10^{-19})(0.4)}{500 \times 10^6}$ = 1.664×10^8 s ≈ 5.28254 years ≈ 5.28 years $\frac{(1.30 \times 10^{30})}{931} \frac{(1.661 \times 10^{-27})(3 \times 10^8)^2}{500 \times 10^6} (0.4) = 5.30$ years	1M 1A 2	1M for $\frac{(1.30 \times 10^{30} \times 10^6)(1.6 \times 10^{-19})}{500 \times 10^6}$ Accept: 5.27 – 5.30 years
(ii) The concentration of uranium-235 nuclei will <u>decrease</u> with time and chain reaction cannot be maintained when the concentration is too low.	1A 1	
(d) (i) moderator: to <u>slow down</u> the fast <u>neutrons</u> produced in fission.	1A 1	
(ii) control rods: to control the rate of nuclear fission / reaction by absorbing the neutrons or for shutting down the reactor in case of emergency.	1A 1	1A: absorbing neutrons + controlling / slowing down the chain reaction Or absorbing neutrons + shutting down the reactor

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(a) The products have a higher binding energy per nucleon and are therefore more stable; the increase in total binding energy is released.

(b)(i) Binding energy is the energy required to separate a nucleus completely into its individual nucleons.

(ii) The product binding energies are $8.27(144) = 1190.88$ MeV and $8.59(90) = 773.10$ MeV. The energy released is $1190.88 + 773.10 - 1783 = 180.98$ MeV ≈ 181 MeV.

(c)(i) The usable energy is $(1.30 \times 10^{30})(1.60 \times 10^{-13})(0.40)$ J. At 500 MW, $t = 1.664 \times 10^8$ s = 5.28 years.

(ii) As U-235 is consumed, its concentration becomes too low to sustain the chain reaction.

(d)(i) A moderator slows fast neutrons so that they are more likely to cause further fission.

(ii) Control rods absorb neutrons and are inserted or withdrawn to regulate or shut down the chain reaction.

Question 4

Section D: Medical Physics

Multiple-choice questions

4.1 When the accommodation of an eye changes from looking at a distant object to a near object, which statements are correct?

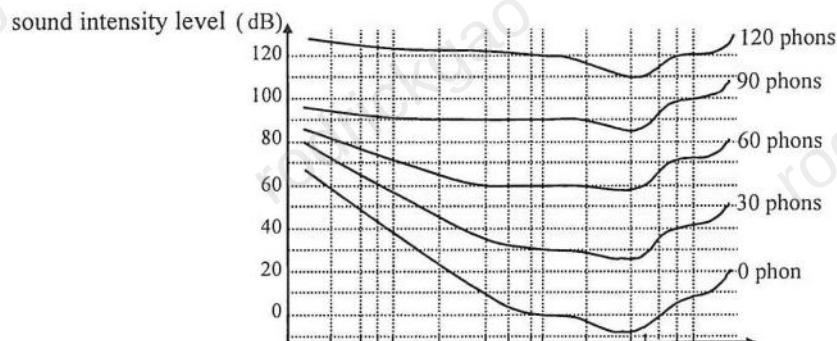
- (1) The eye lens becomes less convex.
- (2) The optical power of the eye lens increases.
- (3) The ciliary muscles around the lens contract.

- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only**

Official answer: D

For near focusing, the ciliary muscles contract, the lens becomes more convex and its optical power increases. Thus (2) and (3) are correct. Answer: D.

4.2 The figure shows the relationship of loudness (in phons) to sound intensity level (in dB) for persons with normal hearing. Which statement CANNOT be deduced from this figure?



- A. The ear is most sensitive to sounds of frequencies between 2000 Hz and 5000 Hz.
- B. A sound of 60 Hz at 40 dB is inaudible.
- C. Hearing sensitivity always increases with the increase of frequency of sound.**
- D. For sounds at frequencies below 1000 Hz, the numerical value of sound intensity level (in dB) is not smaller than its loudness (in phons).

Official answer: C

An equal-loudness curve does not show that hearing sensitivity always increases as frequency rises; sensitivity varies non-monotonically with frequency. Therefore statement C cannot be deduced. Answer: C.

4.3 A doctor suspects that a patient has tumours in the liver. Which methods can be used to detect the tumours and measure their size?

- (1) examining with an endoscope
- (2) taking an ultrasound B-scan
- (3) taking a computed tomography (CT) scan

- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only

Official answer: D

Ultrasound B-scanning and CT can locate and estimate the size of a liver tumour. An endoscope cannot directly reach and image the liver. Thus (2) and (3) are suitable. Answer: D.

4.4 Which statements about the piezoelectric crystal inside an ultrasound transducer are correct?

- (1) The piezoelectric crystal converts electrical signals into mechanical vibrations and vice versa.
- (2) The thickness of the piezoelectric crystal is arbitrary.

- A. Only (1) is correct.
- B. Only (2) is correct.
- C. (1) and (2) are correct.
- D. (1) and (2) are incorrect.

Official answer: A

A piezoelectric crystal converts electrical oscillations to mechanical vibrations and returning vibrations to electrical signals. Its thickness is chosen for resonance, normally related to half a wavelength, rather than arbitrarily. Only statement (1) is correct. Answer: A.

4.5 Using the information below, find the proportion of energy transmitted when ultrasound is incident from air to skin.

Acoustic impedance ($\text{kg m}^{-2} \text{s}^{-1}$): air = 430; soft tissue = 1.5×10^6 .

- A. 5.7×10^{-4}
- B. 1.1×10^{-3}
- C. 2.8×10^{-3}
- D. 1.0×10^{-2}

Official answer: B

The intensity transmission coefficient is $T = 4Z_1Z_2/(Z_1+Z_2)^2$. With $Z_1 = 430$ and $Z_2 = 1.5 \times 10^6 \text{ kg m}^{-2} \text{s}^{-1}$, $T \approx 1.15 \times 10^{-3}$, or 1.1×10^{-3} . Answer: B.

4.6 Which sequence for the procedure of radionuclide imaging (RNI) is correct?

- (1) The drug is moved through the bloodstream to the target organ.
- (2) Inject the drug into the patient's body.
- (3) Label the drug with a radioactive isotope.
- (4) Reconstruct the image by a computer.
- (5) Scan the patient with a gamma camera.

- A. (2) $\rightarrow$ (3) $\rightarrow$ (1) $\rightarrow$ (5) $\rightarrow$ (4)
- B. (2) $\rightarrow$ (3) $\rightarrow$ (1) $\rightarrow$ (4) $\rightarrow$ (5)
- C. (3) $\rightarrow$ (2) $\rightarrow$ (1) $\rightarrow$ (5) $\rightarrow$ (4)
- D. (3) $\rightarrow$ (2) $\rightarrow$ (1) $\rightarrow$ (4) $\rightarrow$ (5)

Official answer: C

The correct sequence is: label the drug (3), inject it (2), allow it to reach the target through the bloodstream (1), detect the emitted gamma rays (5), and reconstruct the image by computer (4). Answer: C.

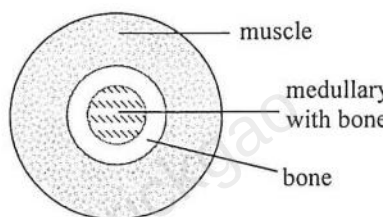
4.7 Radionuclide imaging can be used to study kidney diseases. Which radioactive isotope would be most suitable?

- A. γ radiation; half-life 20.3 minutes
- B. γ radiation; half-life 6.0 hours
- C. β radiation; half-life 2.7 days
- D. β radiation; half-life 3.3 hours

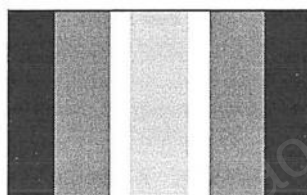
Official answer: B

The tracer should emit penetrating gamma radiation and have a half-life long enough for the procedure but short enough to limit dose. The 6 h gamma emitter best satisfies these requirements. Answer: B.

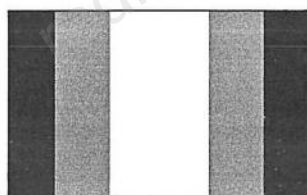
4.8 The figure shows a simplified cross-section of a human arm. The medullary cavity is the central cavity filled with bone marrow. The linear attenuation coefficient of bone marrow is roughly the same as that of muscle. Which figure best represents the X-ray radiographic image of the arm? The four image options A–D are reproduced from the original paper below.



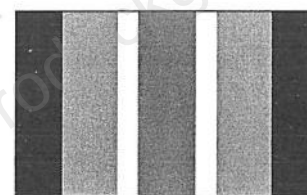
A.



B.



C.



D.



Official answer: A

Bone attenuates X-rays most strongly and therefore appears as bright bands. Marrow has attenuation closer to muscle, so the centre has a shade similar to the surrounding soft tissue. Option A matches this pattern. Answer: A.

Structured question

- (a) State briefly how X-rays are produced. (1 mark)
- (b) The table shows the linear attenuation coefficients of soft tissue and bone for an X-ray beam: soft tissue $\mu_s = 0.51 \text{ cm}^{-1}$; bone $\mu_b = 2.46 \text{ cm}^{-1}$.
- (i) State one factor contributing to a higher linear attenuation coefficient of bone compared with soft tissue. (1 mark)
- (ii) An X-ray beam of intensity I_0 passes through soft tissue 5.6 cm thick and is attenuated to intensity I . What thickness of bone would yield the same degree of attenuation of the same X-ray beam? Show your calculation. (2 marks)
- (iii) Explain why X-ray radiographic imaging of the breast usually employs X-rays of lower energy (about 20 keV), while examination of bony structures employs X-rays of energy around 100 keV. (2 marks)
- (c) People are often concerned about radiation exposure during medical examinations such as X-ray radiographic imaging and computed tomography (CT) scans. The equivalent doses are: taking an X-ray radiographic image, 0.1–0.2 mSv; a CT scan, 1–10 mSv; average weekly natural background dose for a person, about 0.05 mSv.
- (i) State one potential hazard of exposure to ionizing radiation on human bodies. (1 mark)
- (ii) Explain why the equivalent dose of a CT scan is higher than that of taking an X-ray radiographic image. (2 marks)
- (iii) Name one source contributing to the natural background dose. (1 mark)

Official marking scheme

3. D (70)	3. C (70)	1. D (70)	3. A (70)
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Solution	Marks	Remarks
4. (a) X-rays are produced when high-speed electrons strike a heavy metal target (like tungsten).	1A	
	1	
(b) (i) density of bone is higher / bone contains element(s) with high atomic number / heavy elements like calcium in bone block X-rays. (Accept other reasonable answers.)	1A	
	1	
(ii) $I = I_0 e^{-\mu_s t_s} = I_0 e^{-0.51 \times 5.6}$ $I = I_0 e^{-\mu_b t_b} = I_0 e^{-2.46 \times t_b}$ $t_b = 1.16097561 \text{ cm} \approx 1.2 \text{ cm}$ (2 sig. fig.)	1M	Alternative method: $0.51 \times 5.6 = 2.46 \times t_b$
	1A	Accept 1.16 cm to 1.2 cm
	2	
(iii) Imaging of the breast only involves soft tissues and X-rays of longer wavelength / lower frequency / less penetrating power are needed or giving better contrast of soft tissues / more sensitive to small changes in density of soft tissues. For structures with bone, X-rays of shorter wavelength / higher frequency / greater penetrating power are needed.	1A	Accept less harmful with X-rays of lower energy
	1A	
	2	
(c) (i) Induce (cause) cancer / heritable mutations / genetically associated diseases. (Accept other reasonable answers.)	1A	
	1	
(ii) A CT scan delivers a 2-D cross-sectional image of the body part by means of multiple exposure (using a rotating X-ray source(s)), equivalent dose is larger due to a relatively longer time of exposure.	1A	
	1A	
	2	
(iii) cosmic rays / radon gas (from building) / radioactive substances in soil / rock / food or water. (Accept other reasonable answers.)	1A	
	1	

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(a) High-speed electrons strike a heavy-metal target to produce X-rays.

(b)(i) Bone is denser and contains higher-atomic-number calcium, so it attenuates X-rays more strongly than soft tissue.

(ii) Equal attenuation requires $\mu_{\text{muscle}}(5.6) = \mu_{\text{bone}} t$. Thus $t = 0.51(5.6)/2.46 = 1.16 \text{ cm} \approx 1.2 \text{ cm}$.

(iii) Lower-energy X-rays give better soft-tissue contrast and reduce unnecessary penetrating exposure, whereas higher-energy X-rays are needed to penetrate bone.

(c)(i) Ionizing radiation may cause cell damage, cancer or genetic mutation.

(ii) CT takes many projections while the source rotates around the patient, so the exposure and dose are greater than for one radiograph.

(iii) One natural background source is cosmic radiation, radon gas or naturally occurring radionuclides in rocks and food.